

1 Solution — GIAM 2.2

Problem 4

1.1 Problem

Determine a sentence using the *and* connector ($\wedge$) that gives the negation of $A \Rightarrow B$.

1.2 Answer

$$\neg(A \Rightarrow B) \equiv A \wedge \neg B.$$

“It is *not* the case that A implies B ” means exactly “ A is true **and** B is false.”

1.3 Derivation

Start from the material-implication rewrite of Problem 2, $A \Rightarrow B \equiv \neg A \vee B$, negate, and apply **De Morgan’s law** ($\neg(P \vee Q) \equiv \neg P \wedge \neg Q$):

$$\neg(A \Rightarrow B) \equiv \neg(\neg A \vee B) \equiv \neg(\neg A) \wedge \neg B \equiv A \wedge \neg B.$$

1.4 Verification by Truth Table

A	B	$\neg B$	$A \Rightarrow B$	$\neg(A \Rightarrow B)$	$A \wedge \neg B$
T	T	F	T	F	F
T	F	T	F	T	T
F	T	F	T	F	F
F	F	T	T	F	F

Table 1.1

$\neg(A \Rightarrow B)$ is true only when the conditional is broken: A holds but B fails.

A	B	$\neg B$	$A \Rightarrow B$	$\neg(A \Rightarrow B)$	$A \wedge \neg B$
T	T	F	T	F	F
T	F	T	F	T	T
F	T	F	T	F	F
F	F	T	T	F	F

The two blue/brown columns match: the only true row is $A = T, B = F$ (highlighted).

$$\neg(A \Rightarrow B) \equiv A \wedge \neg B$$

Figure 1.1: A truth table with columns A , B , not- B , A implies B , not- $(A$ implies $B)$, and A and not- B . The column not- $(A$ implies $B)$ and the column A and not- B are identical: both false in rows TT, FT, FF, and both true only in the highlighted row $A=T, B=F$. Boxed conclusion: not- $(A$ implies $B)$ is equivalent to A and not- B .

The columns for $\neg(A \Rightarrow B)$ and $A \wedge \neg B$ match in all four rows. Both are true in exactly **one** row — A true, B false. ■

1.5 Remark — The Negation Pinpoints the One Broken Row

A conditional $A \Rightarrow B$ is false in only a single situation: hypothesis true, conclusion false. Its negation must therefore be true in *exactly* that one situation — and $A \wedge \neg B$ is the sentence that “fires” precisely there. This is the quickest way to remember the identity: to *break* the promise “if A then B ,” you need A to actually happen while B fails to. No other row can be a counterexample, which is why the negation is a single conjunction rather than something more complicated.

1.6 Remark — This Is How You Disprove an Implication

The identity is the logical backbone of **finding a counterexample**. To refute a claim “ $A \Rightarrow B$ ” — for instance “if n is prime, then n is odd” — you do *not* argue about implic-

ations directly; you produce a single case where the hypothesis holds and the conclusion fails: $n = 2$ is prime (A) and even ($\neg B$). That witness is exactly an instance of $A \wedge \neg B$. So “the negation of a conditional is a conjunction” is the precise reason a lone counterexample suffices to kill a universally-quantified “if-then” statement — a theme that returns when Chapter 2 negates quantifiers.

1.7 Remark — Note What the Negation Is *Not*

Two tempting wrong answers are worth ruling out. The negation of $A \Rightarrow B$ is **not** $A \Rightarrow \neg B$ (“if A then not B ”): that is a *different, stronger* claim, and it is even *true* in the rows where A is false, so it cannot be the negation (a statement and its negation must disagree in every row). Nor is it $\neg A \Rightarrow \neg B$ or the converse $B \Rightarrow A$. The negation is not a conditional at all — it is a flat conjunction of facts, $A \wedge \neg B$, asserting *both* that A holds *and* that B does not. Dropping the arrow entirely is the whole point.

1.8 Remark — The Companion Negations

The same De Morgan machinery negates the other connectives, and it is useful to see $A \wedge \neg B$ sit among them:

statement	its negation
$A \wedge B$	$\neg A \vee \neg B$
$A \vee B$	$\neg A \wedge \neg B$
$A \Rightarrow B$	$A \wedge \neg B$
$A \Leftrightarrow B$	$A \oplus B$ (exactly one holds)

Table 1.2

Notice the pattern: negation swaps $\wedge \leftrightarrow \vee$ and flips each piece — and a conditional, once unmasked as $\neg A \vee B$, follows the same rule to become $A \wedge \neg B$. This closes a small loop with the very first problem of the chapter: the negation of a biconditional is the **exclusive-or** of Problem 2.1.1, so the XOR circuit built there computes $\neg(A \Leftrightarrow B)$.